SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 5

Strategic Capacity Planning

Capacity, best operating level, utilisation, scale economies, focus and flexibility, requirements, cushions, chunks, system balance, decision trees, service capacity (~70%), VRIN and dynamic capabilities. Case: Pelindo & Semarang Container Terminal, including your syndicate's post-assignment.

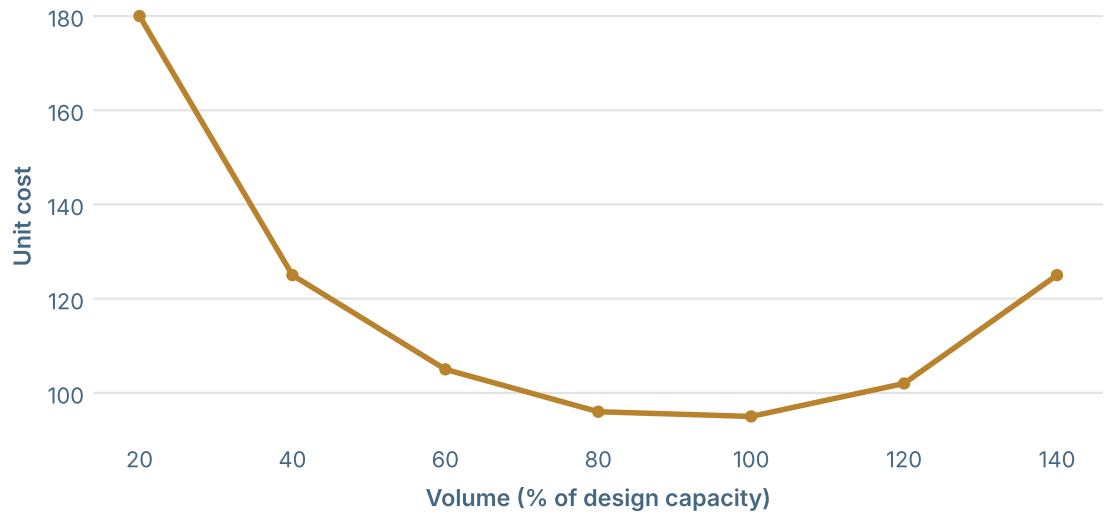
Material

1 Capacity as a strategic commitment

Capacity is a rate over a stated period. Long lead times, irreversibility and asymmetric errors make it strategic. Best operating level = minimum average unit cost; utilisation = capacity used ÷ BOL (480/500 = 96%).

2 The U-shaped unit-cost curve and best operating level Deep dive

Average unit cost vs volume (conceptual)



The lowest point is the best operating level.

3 Scale, focus and flexibility

Economies vs diseconomies of scale; transport often sets plant size; scale ≠ productivity. Focused factories and plant-within-a-plant. Flexible plants, processes (economies of scope) and workers cost money up front.

4 Planning capacity: cushion, chunk, balance

Forecast → convert → project → cushion → chunk → balance. Cushion is a decision ($10m demand on $12m capacity = 20%; reverse = −16.7%). Small frequent vs large infrequent additions trade idle cost against shortage risk and scale. Add capacity at the bottleneck.

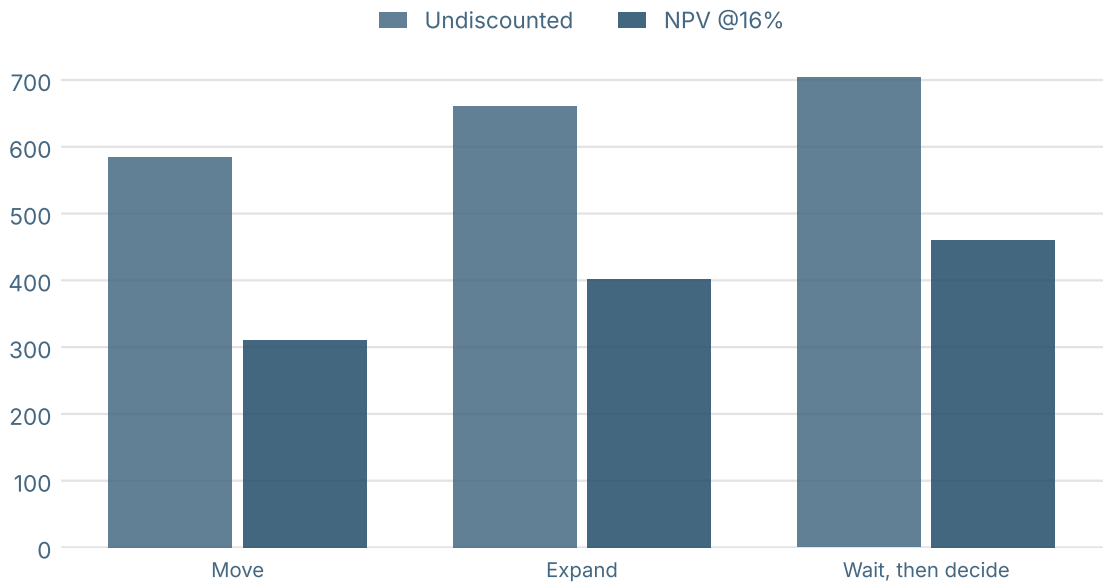
5 Decision trees for capacity alternatives

Squares are decisions, circles chance events; solve right to left. Hackers: move ≈ $585k, expand ≈ $660k, wait-then-decide ≈ $704k — start small, expand if demand is high; at 16% discounting the ranking holds but margins compress.

6 Hackers' decision tree: before and after discounting

Deep dive

Hackers: expected values ($000)



The ranking holds; start small, expand if demand is high. Discounting cuts values 35–47%.

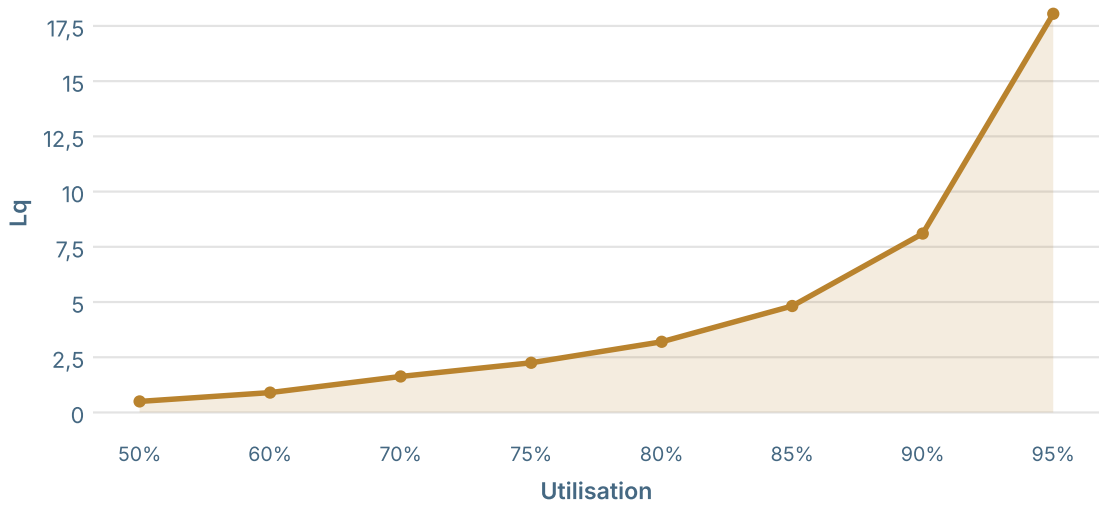
7 Service capacity is different

Service capacity is perishable, must be located near customers, faces volatile demand, is planned in minutes and targets ~70% utilisation; beyond, quality drops before cost. Targets vary (ER low, commuter rail near 100%).

8 Why services target ~70%: queues explode at high utilisation

Deep dive

Queue length vs utilisation (single server)



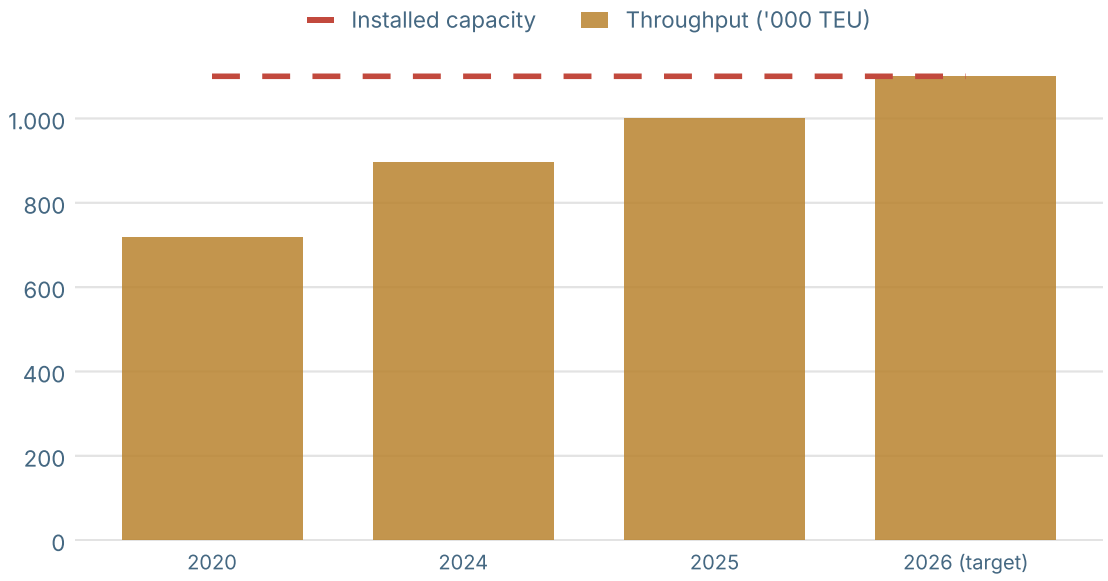
Lq at 85% is ~3× that at 70%; at 95% more than 10×.

9 RBV/VRIN and dynamic capabilities

VRIN: valuable → rare → inimitable → non-substitutable (+ organised). Buyable capacity gives parity. Five Forces, RBV and dynamic capabilities ask complementary questions; sense, seize, transform — convert before you duplicate.

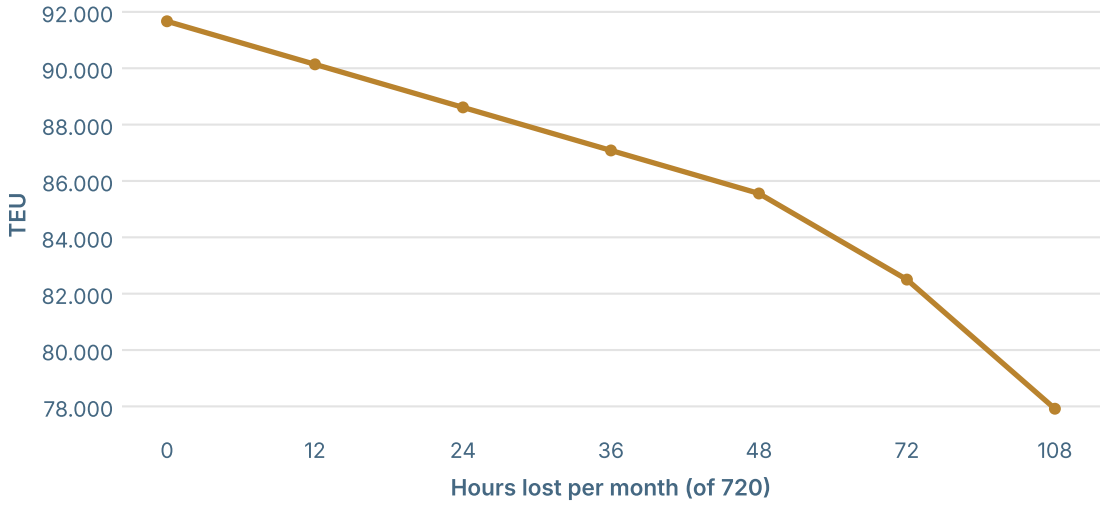
10 Semarang terminal in charts Deep dive

TPKS throughput ('000 TEU) and utilisation



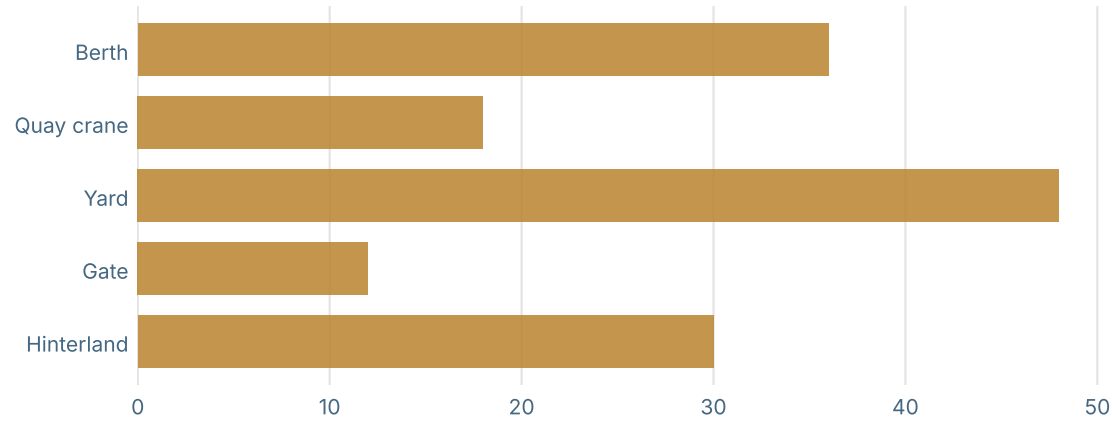
Utilisation 65.2% → 81.4% → ~90.9% → 100%.

Effective monthly capacity vs flood hours lost



Every 36 hours of monthly downtime removes ~4,600 TEU of effective capacity.

Investment lead time by stage (months, midpoint)



Midpoints of the case ranges.

Key takeaways

1. Capacity is a rate.
2. Utilisation = used ÷ BOL.
3. The cushion is a decision.
4. Add at the bottleneck.
5. Solve trees right to left.
6. Services target ~70%.
7. Transform before you duplicate.

Case studies

Case & worked answers

Pelindo & Semarang Container Terminal: capacity on sinking ground

Throughput grew to >1m TEU on 1.1m capacity, giving a zero 2026 cushion; subsidence >12 cm/yr and a May 2025 flood halted operations. Q1: annual utilisation hides peaks — measure weekly YOR, dwell time, vessel waits. Q2: the yard is the least elastic stage; cranes alone move the queue ashore; the hinterland may bind. Q3: effective capacity = installed × net available operating

hours index (720 − 36)/720 = 95% → ~87,083 TEU/month. Q4: full B is irreversible and risks stranding; the hybrid has the best EV/downside. Q5: the Tanjung Emas site is VRIN yet caps capacity. Your syndicate's recommendation: sequential hybrid with Chunk 1 (US$8–12m), Chunk 2 dry-port JV (US$20–25m), YOR ≥85% trigger and subsidence ≥15 cm or downtime ≥10% kill criteria. Improvement: quantify probabilities and EVs.

Worked examples

Problem 1 **Easy** **Utilisasi dan cushion**

Lini produksi: best operating level 100 unit/jam, output aktual 85 unit/jam. Pabrik lain: kapasitas 1.200 unit/hari, permintaan 1.000 unit/hari. Hitung utilisasi lini pertama dan capacity cushion pabrik kedua.

1

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Utilisasi = Kapasitas terpakai ÷ Best operating level = 85 ÷ 100 = 85%
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Cushion = (Kapasitas − Permintaan) ÷ Permintaan = 200 ÷ 1.000 = 20%
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Answer: Utilisasi 85%; cushion 20%.

Problem 2 Medium **Kapasitas efektif saat banjir**

Kapasitas terpasang 100.000 TEU/bulan pada 720 jam terjadwal. Bulan ini 54 jam hilang karena banjir rob. Berapa kapasitas efektif?

1. Index = (720 − 54) ÷ 720 = 0,925.
2. Kapasitas efektif = 100.000 × 0,925 = 92.500 TEU.

Answer: 92.500 TEU (hilang 7.500 TEU).

Problem 3 Medium **Decision tree ekspansi**

Permintaan tinggi p = 0,6, rendah 0,4. Ekspansi besar: 500 / −100. Ekspansi kecil: 300 / 100 (Rp juta). Mana yang dipilih?

1. EV besar = 0,6 × 500 + 0,4 × (−100) = 300 − 40 = 260.
2. EV kecil = 0,6 × 300 + 0,4 × 100 = 180 + 40 = 220.
3. Besar unggul 40, tetapi berisiko rugi 100 jika permintaan rendah.

Answer: Ekspansi besar (EV 260 vs 220) — kecuali perusahaan tidak sanggup menanggung kemungkinan rugi Rp100 juta.

Practice questions

Try answering first. The answer key is below each question.

1. Capacity 1.1m TEU, throughput 1.0m. Utilisation…

A. 81%
B. 90.9%
C. 100%
D. 110%

Answer: B. 1.0/1.1.

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2. Adding cranes when the yard is full…

A. Lifts throughput
B. Moves the queue ashore
C. Cuts dwell time
D. Reduces floods

Answer: B. Add at the bottleneck.

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3. Face-to-face services typically target about…

A. 40%
B. 70%
C. 95%
D. 100%

Answer: B. Above that, queues grow.

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4. A resource anyone can buy (cranes) gives, per VRIN…

A. Sustained advantage
B. Parity
C. Temporary advantage
D. Disadvantage

Answer: B. Valuable but not rare.

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5. 'Transforming' is best illustrated by…

A. Building a duplicate plant
B. Converting an underused gas line to oil
C. Overtime
D. Raising prices

Answer: B. Energy East.

Sources

- [STS] Strategic Capacity Planning (MK4)
- Jacobs & Chase, Bab 5
- Pelindo & Semarang Container Terminal case (SBM ITB)
- Syndicate 3 — Post-Assignment MK4 Week 5 (tugas kelompokmu)